

RAJALAKSHMI ENGINEERING COLLEGE

Experiment 3: EDA - Data Cleaning

Name	Kabilan S	Register No.	241501502
Department	AIML	Year / Semester	III / V

AIM

To clean a loan dataset by identifying missing values, visualizing missingness, imputing values, removing duplicates, standardizing categories, and scaling numeric data.

SOURCE

EXP3/Expt-03.ipynb

PYTHON CODE - CELL 1

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler, StandardScaler
```

PYTHON CODE - CELL 2

```
df = pd.read_csv("data.csv")

print("Initial Data Overview")
print(df.info())
```

EXECUTED OUTPUT

```
Initial Data Overview
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Loan_ID                7 non-null      object
1   Gender                 5 non-null      object
2   Married                6 non-null      object
3   Dependents             6 non-null      object
4   Education              7 non-null      object
5   Self_Employed          6 non-null      object
6   ApplicantIncome        7 non-null      int64
7   CoapplicantIncome      7 non-null      int64
8   LoanAmount             6 non-null      float64
9   Loan_Amount_Term       6 non-null      float64
10  Credit_History          6 non-null      float64
11  Property_Area          7 non-null      object
12  Loan_Status            7 non-null      object
dtypes: float64(3), int64(2), object(8)
memory usage: 860.0+ bytes
None
```

PYTHON CODE - CELL 3

```
print("\nMissing Values:")
print(df.isnull().sum())
```

EXECUTED OUTPUT

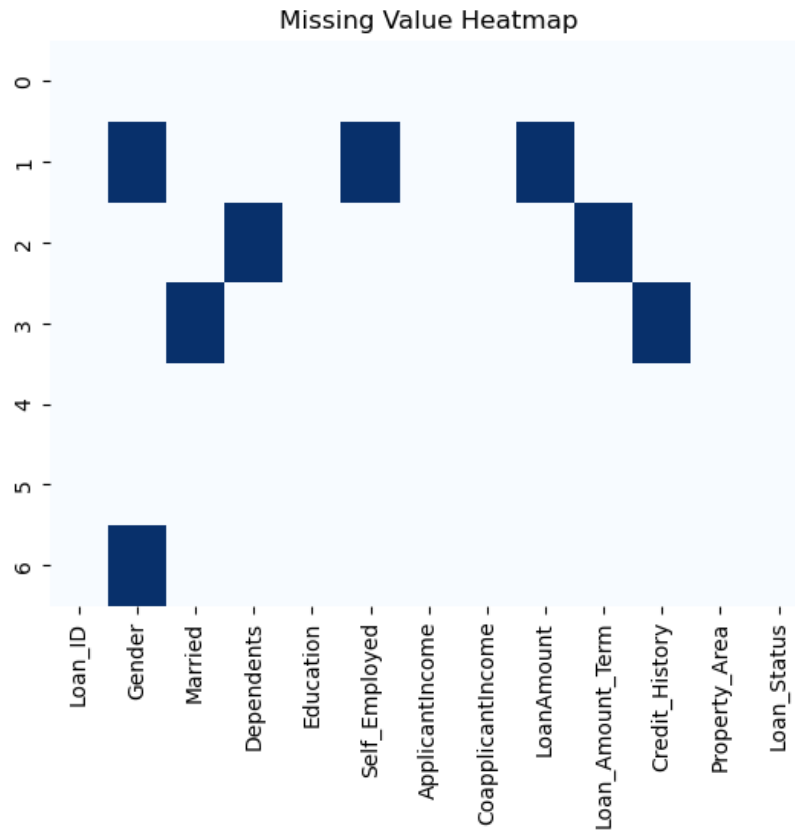
```
Missing Values:
Loan_ID                0
Gender                 2
Married                1
Dependents             1
Education              0
Self_Employed          1
ApplicantIncome        0
CoapplicantIncome      0
LoanAmount             1
Loan_Amount_Term       1
```

```
Credit_History      1
Property_Area       0
Loan_Status         0
dtype: int64
```

PYTHON CODE - CELL 4

```
sns.heatmap(df.isnull(), cbar=False, cmap="Blues")
plt.title("Missing Value Heatmap")
plt.show()
```

EXECUTED OUTPUT



PYTHON CODE - CELL 5

```
for col in ["Gender", "Married", "Dependents", "Self_Employed"]:
    df[col] = df[col].fillna(df[col].mode()[0])

df["LoanAmount"] = df["LoanAmount"].fillna(df["LoanAmount"].median())
df["Loan_Amount_Term"] = df["Loan_Amount_Term"].fillna(df["Loan_Amount_Term"].mode()[0])
df["Credit_History"] = df["Credit_History"].fillna(df["Credit_History"].mode()[0])

initial_rows = len(df)
df.drop_duplicates(inplace=True)
print("\nDuplicates Removed:", initial_rows - len(df))
```

EXECUTED OUTPUT

```
Duplicates Removed: 1
```

PYTHON CODE - CELL 6

```
df["Dependents"] = df["Dependents"].replace("3+", 3).astype(int)

for col in ["Gender", "Married", "Education", "Self_Employed", "Property_Area"]:
    df[col] = df[col].str.strip().str.capitalize()

minmax = MinMaxScaler()
df[["ApplicantIncome", "CoapplicantIncome", "LoanAmount"]] = minmax.fit_transform(
    df[["ApplicantIncome", "CoapplicantIncome", "LoanAmount"]]
)

standard = StandardScaler()
```

```
df[["Credit_History"]] = standard.fit_transform(df[["Credit_History"]])

print("\nCleaned Data")
print(df.info())
print(df.head())
```

EXECUTED OUTPUT

```
Cleaned Data
<class 'pandas.core.frame.DataFrame'>
Index: 6 entries, 0 to 5
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Loan_ID                6 non-null     object
1   Gender                 6 non-null     object
2   Married                6 non-null     object
3   Dependents             6 non-null     int32
4   Education              6 non-null     object
5   Self_Employed          6 non-null     object
6   ApplicantIncome        6 non-null     float64
7   CoapplicantIncome      6 non-null     float64
8   LoanAmount             6 non-null     float64
9   LoanAmount_Term        6 non-null     float64
10  Credit_History          6 non-null     float64
11  Property_Area          6 non-null     object
12  Loan_Status            6 non-null     object
dtypes: float64(5), int32(1), object(7)
memory usage: 648.0+ bytes
None
   Loan_ID  Gender  Married  Dependents  Education  Self_Employed  \
0  LP001    Male    Yes         0      Graduate          No
1  LP002  Female    Yes         3      Graduate          No
2  LP003  Female    No         0  Not graduate          Yes
3  LP004    Male    Yes         1      Graduate          No
4  LP005  Female    Yes         2      Graduate          No

   ApplicantIncome  CoapplicantIncome  LoanAmount  LoanAmount_Term  \
0           1.000000           0.000000         1.0000           360.0
1           0.555634           0.359390         0.7875           360.0
2           0.000000           0.000000         0.0000           360.0
3           0.848368           1.000000         0.7750           360.0
4           0.351000           0.595806         0.8000           360.0

   Credit_History  Property_Area  Loan_Status
0           0.447214          Urban           Y
1           0.447214          Urban           N
2          -2.236068          Rural           Y
3           0.447214          Urban           Y
4           0.447214       Semiurban           Y
```

RESULT

Missing values and duplicates were handled, categorical values were standardized, and numeric columns were scaled successfully, as confirmed by the cleaned-data output.